PHASE 4.2.3 ABSOLUTE PROTOCOL FREEZE

SimuPharm Educational Pilot Evaluation

Final Technical Sanity Audit, Statistical Methodology Correction & Governance Certification

Date: October 4, 2026
Governance Status: NO-GO — ETHICS PENDING
Engine Status: MATCH — FROZEN

1. EXECUTIVE SUMMARY & STUDY ARCHITECTURE LOCK

This audit constitutes the definitive, immutable pre-recruitment certification for the **SimuPharm Educational Evaluation Study**. The study architecture is completely **LOCKED**: single-group within-subjects pre/post exploratory pilot ($N \approx 25$ undergraduate pharmacy and chemistry students), standardized 50-minute scheduled laboratory session, 10-item isomorphic diagnostic knowledge tests, 4-scenario troubleshooting task evaluated via a 24-point analytical rubric, and the canonical Brooke (1996) System Usability Scale (SUS). **Zero software modifications have been made to the simulation engine, no study arms or participant-facing features have been added, no participant data has been collected or fabricated, and all analysis templates remain 100% blank.**

2. CORRECTIONS IMPLEMENTED FROM PHASE 4.2.2 AUDIT

#	Methodological Domain	Phase 4.2.2 Baseline Finding	Phase 4.2.3 Precise Calibration	Audit Status
1	**Readiness Language**	Stated: "Gate 1 (Scientific & Statistical Readiness): CLEARED (100%)", inadvertently implying absolute pre-empirical readiness.	Replaced with: **"Technical and statistical protocol verification: COMPLETED"**. All pre-data "100% readiness" claims removed.	CORRECTED
2	**Hodges–Lehmann CI Terminology**	Previously characterized as an "Exact Inversion CI", while the critical index k_{crit} was calculated using a normal z approximation.	Designated precisely as: **"Wilcoxon/Walsh order-statistic confidence interval using the prespecified critical-index approximation"**. Word "exact" removed.	CORRECTED
3	**Rubric Guessing Hypothesis**	Included a one-sample inferential hypothesis test: $H_0: \text{Median Rubric} \le 6.0/24$, arbitrarily assuming 6 points as "random guessing".	**Excised inferential test.** The multi-component rubric (ID, reasoning, action) lacks a mathematical guessing model; retained purely as a descriptive outcome.	CORRECTED
4	**SUS Usability Framework**	Posed a one-sample Wilcoxon test against the 68.0 benchmark ($H_0: \text{Median SUS} \le 68.0$), blurring benchmarking with inferential claims.	**Excised inferential test.** Reclassified as **contextual descriptive benchmarking** (mean, median, SD, IQR); benchmark is not a control group or proof of efficacy.	CORRECTED
5	**Qualitative Paradigm Consistency**	Incompatibly combined reflexive thematic analysis (Braun & Clarke 2006) with dual coding and a mandatory inter-coder threshold ($\kappa \ge 0.80$).	Adopted a unified framework: **Structured thematic analysis with an inductively developed codebook, 25% dual coding, descriptive agreement κ, and consensus resolution**.	CORRECTED
6	**Internal-Consistency Reliability**	Mentioned Cronbach's α for the 10-item pre/post test without qualification, risking over-interpretation at pilot sample size ($N \approx 25$).	Explicitly designated as **exploratory and unstable at $N \approx 25$**; not used to establish psychometric validity; tests not claimed as "validated" based on α.	CORRECTED
7	**Hake Normalized Gain Framing**	Risked being perceived as an inferential hypothesis test or basis for categorizing individual students into learning brackets.	Explicitly stated: **"The normalized gain is reported descriptively and is not used as evidence of causal educational effectiveness."** Individual tagging removed.	CORRECTED
8	**Multiplicity & Family Definition**	Holm-Bonferroni correction was previously applied across a mixed bag of hypothesis tests and descriptive metrics ($k=2$ to $k=5$).	With secondary tests reclassified as descriptive benchmarking, **Tier 1 is a standalone primary inferential test ($\alpha = 0.05$)**, eliminating artificial multiplicity penalties.	CORRECTED
9	**Inter-Rater Reliability Metrics**	Grouped Weighted Kappa (κ_w) and $\text{ICC}(2,1)$ without clearly distinguishing their application to rubric measurement levels.	Explicitly distinguished: **κ_w for component-level ordinal agreement (0–2)** vs **$\text{ICC}(2,1)$ for grand-total score reliability (0–24)**.	CORRECTED
10	**Study Allocation Terminology**	Referred to "randomized Participant IDs", risking confusion with a randomized controlled trial design.	Replaced with: **"assigned anonymous Participant IDs (`P01` to `P25`)"** across all intervention and ethics documents.	CORRECTED

3. HODGES–LEHMANN CONFIDENCE INTERVAL & EFFECT-SIZE VERIFICATION

Walsh Order-Statistic CI (Normal-Approximation)

For N complete-case pairs, $M = \frac{N(N+1)}{2}$ pairwise Walsh averages $A_{(k)} = \frac{d_i + d_j}{2}$ ($1 \le i \le j \le N$) are calculated and sorted in ascending order.

- **Point Estimator:** $\text{HL} = \text{Median}(\{A_{(k)}\})$.
- **Critical Index Approximation (Bauer 1972):** The bounds are order statistics $A_{(k_{\text{crit}})}$ and $A_{(M - k_{\text{crit}} + 1)}$, where: $$k_{\text{crit}} = \left\lfloor \frac{N(N+1)}{4} - z_{1-\alpha/2}\sqrt{\frac{N(N+1)(2N+1)}{24}} \right\rfloor$$
- **Epistemological Clarification:** Because k_{crit} relies on the continuous normal approximation of the discrete Wilcoxon distribution, this interval is **NOT called an "exact" interval**; it is a *Walsh order-statistic confidence interval using the prespecified critical-index approximation*. For $N=25$, $M=325$, $k_{\text{crit}}=89$, yielding $[A_{(89)}, A_{(237)})]$.

Signed Rank-Biserial Correlation (r_{rb})

Non-parametric effect size is computed directly from Pratt rank sums of positive (W^+) and negative (W^-) differences:

- **Formula:** $r_{rb} = \frac{W^+ - W^-}{W^+ + W^-} \quad (\text{if } W^+ + W^- = 0, \ r_{rb} = 0.0)$.
- **Directional Verification across 7 Stress Cases:**
  - Case A (All Positive): $r_{rb} = +1.000, \ p = 0.0078$ PASS
  - Case B (All Negative): $r_{rb} = -1.000, \ p = 0.0078$ PASS
  - Case C (Mixed Ranks): $r_{rb} = +0.722, \ p = 0.0781$ PASS
  - Case D (Pratt Zeros): $r_{rb} = +1.000, \ p = 0.0474$ PASS
  - Case E (All Zeros): $r_{rb} = 0.000, \ p = 1.0000$ PASS
  - Case F (Balanced Ties): $r_{rb} = 0.000, \ p = 1.0000$ PASS
  - Case G (Missing Pairs): Complete-case filtering ($N_{\text{valid}}=4$) PASS

4. FINAL INFERENTIAL HIERARCHY & PRE-REGISTRATION ARCHITECTURE

Following the removal of ungrounded secondary one-sample tests, the outcome structure is finalized into an uncompromised, pre-registered three-tier hierarchy:

TIER 1: PRIMARY PRESPECIFIED INFERENTIAL ANALYSIS (STANDALONE $\alpha = 0.05$, TWO-TAILED)

• **Paired Troubleshooting Score Gain ($\Delta\text{Score} = \text{PostTest} - \text{PreTest}$):** Two-tailed Wilcoxon Signed-Rank Test using Pratt's method (`zero_method='pratt'`). Reported with Hodges–Lehmann median difference with 95% Walsh order-statistic CI (critical-index approximation) and signed rank-biserial correlation (r_{rb}). Paired Student's t-test pre-specified as sensitivity analysis.

TIER 2: SECONDARY INFERENTIAL HYPOTHESIS TESTS

• **Status: NONE PRE-SPECIFIED.** Because the 24-point practical rubric lacks a mathematical guessing model and SUS is a subjective usability questionnaire, these metrics are properly evaluated via descriptive benchmarking rather than artificial one-sample null hypotheses. *No family-wise error adjustments or multiplicity penalties are required.*

TIER 3: EXPLORATORY & DESCRIPTIVE OUTCOMES (NO MULTIPLICITY PENALTY, HYPOTHESIS-GENERATING)

• **Practical Rubric Mastery:** Grand score (0–24) and components (ID, Reasoning, Action: 0–2 each); medians, IQRs, percentage of maximum.
• **System Usability Scale (SUS):** Mean, SD, median, and IQR compared contextually against the widely cited 68.0 benchmark.
• **Diagnostic Latency:** Median Time-to-Diagnosis in seconds with IQR across practical scenarios.
• **Exploratory Confidence–Performance Association:** Spearman ρ and Pearson r between confidence ratings (1–5) and rubric performance (zero causal claims).
• **Hake Normalized Gain ($\langle g \rangle$):** Cohort average normalized gain reported descriptively without individual student categorization.
• **Exploratory Item Analysis:** Item difficulty (P) and discrimination (D) with explicit small-sample ($N \approx 25$) stability caveats.
• **Subgroup Summaries:** B.Pharm vs Chemistry; prior HPLC experience vs novices (descriptive medians/IQRs only; zero inferential claims).

5. PRACTICAL RUBRIC & SUS METHODOLOGICAL JUSTIFICATIONS

24-Point Analytical Rubric Justification

The 4-scenario practical troubleshooting task evaluates three distinct cognitive/procedural dimensions: Symptom ID (0–2), Physical Reasoning (0–2), and Corrective Action Plan (0–2), yielding 0–6 per scenario and 0–24 grand total.

- **Rejection of Arbitrary Guessing Null:** Multiple-choice guessing ($p=0.25$) does not apply to open-response rubric scoring. Setting $H_0 \le 6.0/24$ arbitrarily assumes that a student guessing randomly scores exactly 1.5 points per task, which is unsupported by psychometric theory.
- **Reporting Standard:** Evaluated descriptively via median, IQR, component distributions, and scenario-level difficulty rankings.

SUS Contextual Benchmarking Framework

The 10-item System Usability Scale (Brooke 1996) is scored continuously (0–100) using Sauro & Lewis (2016) item imputation for records with ≤ 2 missing items.

- **Benchmarking vs Hypothesis Testing:** The 68.0 benchmark (Bangor et al., 2008) is an empirical industry average derived across hundreds of diverse software systems. It is NOT an experimental control group.
- **Boundary of Interpretation:** A cohort mean or median ≥ 68.0 indicates acceptable usability contextually; it does **NOT prove educational effectiveness or simulator superiority**.

6. QUALITATIVE & RELIABILITY METHODOLOGICAL CONSISTENCY

Structured Qualitative Framework

To resolve paradigm conflicts, the qualitative methodology is formalized as *structured thematic analysis with an inductively developed codebook*:

- **Dual-Coding Protocol:** Two independent researchers code a random 25% sample ($n \approx 6\text{--}7$ transcripts) across three themes (Visual Cues, Misconceptions, Usability).
- **Inter-Coder Agreement (κ):** Cohen's Kappa is reported descriptively as an agreement statistic on coded segments. It is **NOT claimed to mathematically validate the qualitative findings**.
- **Consensus Adjudication:** Discordant codes are reconciled by discussion before coding the remaining 75% of transcripts.

Inter-Rater Rubric Reliability Metrics

Two faculty raters independently score all participant response sheets blind to participant identity. Inter-rater reliability is partitioned by measurement level:

- **Component-Level Ordinal Agreement:** Weighted Cohen's Kappa (κ_w, linear weighting) evaluated on individual 0–2 item ratings (Fault ID, Reasoning, Action Plan).
- **Grand-Total Reliability:** Intraclass Correlation Coefficient $\text{ICC}(2,1)$ (two-way random effects, single rater, absolute agreement) evaluated on the 0–24 aggregate score.
- **Adjudication:** Discrepancies >1 point are resolved via consensus to yield `task_rubric_grand_total`.

7. PSYCHOMETRIC & HAKE-GAIN METHODOLOGICAL LIMITATIONS

Item-Level Psychometric Caveats

- **Sample Size Instability:** For $N \approx 25$, Item Difficulty (P) and Item Discrimination (D) indices describe relative item behavior in this pilot cohort only; they do not represent population parameters.
- **Internal Consistency (Cronbach's α):** Classical test reliability estimates with 10 items and $N \approx 25$ have wide confidence intervals and are unstable. Internal consistency is reported as exploratory; **the pre/post tests are NOT claimed as "psychometrically validated" based on α**.

Hake Normalized Gain Boundaries

- **Descriptive Function:** Normalized gain $\langle g \rangle = \frac{\bar{x}_{\text{post}} - \bar{x}_{\text{pre}}}{10 - \bar{x}_{\text{pre}}}$ describes the proportion of possible gain achieved relative to cohort baseline.
- **Non-Causal Restriction:** Explicitly reported descriptively and **NOT used as evidence of causal educational effectiveness**.
- **No Individual Categorization:** Students are not classified into arbitrary "low/medium/high learning" bins; only cohort distribution and aggregate values are reported.

8. FINAL CLAIM-LANGUAGE AUDIT REGISTER (12 PRESCRIBED PHRASES)

Prescribed Term	Audit Status	Context in Study Documents & Codebases	Governance Action Taken
"100%"	SAFE	Used only for technical facts: simulator runs "100% offline", engine is "100% frozen", template has "0 data rows".	Excised from Gate 1 readiness language ("CLEARED [100%]" removed).
"confirmatory"	SAFE	Appears only in historic audit changelogs as a past finding that was resolved.	Replaced with "Primary Prespecified Inferential Analysis" across all active SAP and RQ files.
"validated"	SAFE	Appears in Prohibited Claims ledger ("Validated across 300+ APIs") and as a limitation warning.	Preserved in Prohibited Claims ledger. Pre/post tests explicitly not claimed as validated based on α.
"psychometrically validated"	SAFE	Zero positive occurrences in active methodology or claims documents.	Confirmed absent.
"random guessing threshold"	SAFE	Mentioned only to disclaim it ("No arbitrary 6/24 random guessing threshold is assumed").	Preserved as an explicit methodological disclaimer.
"proves"	SAFE	Appears only in Prohibited Claims register ("Proves SimuPharm replaces physical wet laboratory").	Calibrated to "indicates" in answer key. Preserved in Prohibited Claims.
"effective"	SAFE	Zero claims asserting proven instructional effectiveness.	Confirmed absent.
"superior"	SAFE	Zero claims asserting superiority over traditional wet laboratory training.	Confirmed absent.
"causes"	SAFE	Used exclusively in multiple-choice distractor options explaining physical HPLC mechanisms.	Preserved in assessment instruments. No causal study design claims made.
"exact CI"	SAFE	Zero occurrences in active methodology files.	Replaced with "Walsh order-statistic CI using prespecified critical-index approximation".
"randomized"	SAFE	Removed from participant tracking descriptions.	Replaced with "assigned anonymous Participant IDs" (`P01` to `P25`).
"generalizable"	SAFE	Zero claims of generalizability exist; single-institution pilot boundaries declared.	Confirmed absent.

9. DATA INTEGRITY & CRYPTOGRAPHIC ENGINE VERIFICATION

Data Schema Purity Verification

- **Schema Location:** `05_CORRECTED_DATA_SCHEMA/participant_data_template.csv`
- **Audited Row Count:** Exactly **1 header row** (73 variables), **0 participant data rows**.
- **Fabrication Audit:** Zero participant records, zero fabricated scores, zero synthetic participant IDs presented as empirical data.
- **Isolation:** Stress-test datasets are strictly segregated in `06_CORRECTED_ANALYSIS/test_data/DEMO_TEST_ONLY.csv` .

Simulation Engine Hash Verification

Source File	Calculated SHA-256 Checksum	Verdict
`hplc_simulator_script.js`	`2fad9547f9fc7c671176346e78732be30246a515853631dede21ae8f7d3b7167`	MATCH — FROZEN
`script.js`	`9812fea221fee06391a35d2680c28ce375ca62bc26bc265ed07f36d4d96f0600`	MATCH — FROZEN
`index.html`	`dfb896b41790efcc365a3721b8ccc7aa8f49861485a74b02538eb38cadbdee36`	MATCH — FROZEN
`hplc_simulator.html`	`02b2b226bf291c2466c23832382e296523cecc68811afe26ba65822197afd6bb`	MATCH — FROZEN

10. STUDY LIMITATIONS & RECRUITMENT GATE DETERMINATION

Mandatory Manuscript Disclaimers: (1) *Single-Group Design:* The within-subjects pre/post design cannot establish causal superiority over traditional laboratory instruction. (2) *Short-Term Evaluation:* Evaluates diagnostic gains immediately following a single 50-minute laboratory session; delayed retention is unmeasured. (3) *Transfer to Physical Hardware:* Transfer of diagnostic skills to real physical HPLC instruments remains unmeasured. (4) *Small Sample Size ($N \approx 25$):* Subgroup comparisons (B.Pharm vs Chemistry) are exploratory and underpowered.

FINAL RECRUITMENT READINESS VERDICT: NO-GO — INSTITUTIONAL ETHICS AUTHORIZATION PENDING

Technical & Statistical Protocol Verification: COMPLETED — Code, assessments, rubrics, analysis pipelines, and templates are technically verified.

Institutional Ethics Authorization: PENDING — Written determination from the host institution's Institutional Ethics Committee (IEC / IRB) has not yet been issued.

Governance Mandate: In accordance with international research ethics standards (Declaration of Helsinki) and host-institution regulations, no student participant may be recruited, consented, or assessed until formal written institutional authorization is secured. Zero self-exemption is permitted.

PROTOCOL STATUS:

FROZEN PENDING INSTITUTIONAL ETHICS AUTHORIZATION

NEXT SCIENTIFIC ACTION:

REAL PARTICIPANT DATA COLLECTION AFTER AUTHORIZATION